EduAgent: A Multi-Agent AI Educational Ecosystem with Inter-Agent Collaboration and Adaptive Learning

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Abstract

This paper presents EduAgent, a full-stack multi-agent AI educational platform that provides students with six specialised AI agents sharing a unified knowledge layer comprising a RAG vector store, concept graph, and persistent student state. Unlike single-chatbot solutions, EduAgent employs a LangGraph-based orchestrator that classifies student intent and routes queries to the optimal agent. The platform introduces inter-agent collaboration where agents consult each other's data — weak-point awareness, mastery state, and overdue reviews — to deliver contextually richer responses. Key innovations include SM-2 spaced repetition with auto-generated flashcards, AI-powered personalised study planning, voice-enabled oral viva simulation, multimodal diagram understanding via Gemini, WebSocket-based collaborative study rooms, and a custom agent builder for professors. The system is deployed as a five-service Docker Compose stack (PostgreSQL, Qdrant, Redis, FastAPI, Next.js) with 60+ automated tests. Evaluation through a structured user study protocol demonstrates the platform's effectiveness in improving learning outcomes through adaptive, multi-agent interaction.

Keywords: *Multi-Agent Systems, Educational AI, Retrieval-Augmented Generation, Spaced Repetition, LangGraph, Adaptive Learning, Inter-Agent Collaboration*

I. Introduction

The proliferation of large language models (LLMs) has created new opportunities for AI-assisted education. However, most existing platforms employ a single-agent chatbot paradigm that struggles to differentiate between distinct educational activities such as concept explanation, exam preparation, self-assessment, and progress tracking. Students must context-switch between multiple tools, losing the continuity of their learning state.

EduAgent addresses this gap by implementing a multi-agent architecture where six specialised AI agents — Concept QA, Exam Predictor, Weak-Point Tracker, Protégé (Feynman technique), Oral Viva, and Essay Critique — operate within a shared data layer. A LangGraph-based orchestrator automatically classifies student intent and routes each query to the most appropriate agent, while a unified student state model tracks concept mastery, review schedules, and learning events across all interactions.

This paper makes the following contributions: (1) a multi-agent educational architecture with inter-agent collaboration, where agents consult each other's data before responding; (2) integration of SM-2 spaced repetition with auto-generated flashcards extracted from conversations; (3) an AI-powered study planner that generates personalised schedules based on mastery data; (4) voice-enabled oral examination simulation; (5) multimodal diagram understanding using Gemini's vision capabilities; (6) WebSocket-based collaborative study rooms with a shared AI assistant; (7) a custom agent builder enabling professors to create domain-specific assessment agents; and (8) comprehensive class-level analytics with agent performance monitoring.

II. Related Work

Educational AI systems have evolved from rule-based tutoring systems to LLM-powered chatbots. Khan Academy's Khanmigo [1] provides a single-agent AI tutor but lacks multi-agent specialisation. Duolingo Max [2] uses GPT-4 for language learning but is domain-specific. Carnegie Learning's MATHia [3] employs adaptive algorithms but without LLM-based natural language interaction.

Multi-agent systems have been explored in software engineering (MetaGPT [4], AutoGen [5]) but their application to education remains limited. The concept of spaced repetition (SM-2 algorithm by Wozniak [6]) is well-established in tools like Anki but has not been integrated with conversational AI agents.

Retrieval-Augmented Generation (RAG) [7] enables grounding LLM responses in specific course material. Vector databases like Qdrant provide efficient similarity search for this purpose. LangGraph [8] extends LangChain with stateful graph-based workflows suitable for multi-agent orchestration.

EduAgent differs from prior work by combining multi-agent routing, inter-agent collaboration, spaced repetition, RAG, and real-time collaboration in a unified full-stack platform.

III. System Architecture

EduAgent employs a five-service architecture deployed via Docker Compose:

Frontend (Next.js 16): Server-side rendered React application with TailwindCSS, shadcn/ui components, Zustand for state management, React Flow for concept graph visualisation, and Web Speech API for voice interactions. The frontend communicates with the backend via REST APIs and Server-Sent Events (SSE) for streaming responses, plus WebSocket for real-time study room collaboration.

Backend (FastAPI): Asynchronous Python API server with 15 routers handling authentication, chat, flashcards, analytics, study planning, progress reports, custom agents, study rooms, image analysis, and WebSocket connections. The backend uses SQLAlchemy 2.0 with async PostgreSQL access and Alembic for migrations.

PostgreSQL 16: Relational database storing user accounts, concept graphs with prerequisite edges, student mastery state (SM-2 parameters), conversations, messages, flashcards, agent logs, learning events, study plans, custom agents, and study rooms. The schema spans 14 tables across 3 Alembic migrations.

Qdrant: Vector database storing sentence-transformer embeddings (all-MiniLM-L6-v2, 384 dimensions) of ingested course material, enabling semantic retrieval for RAG. Collections are partitioned by subject.

Redis 7: In-memory cache for frequently accessed analytics queries and RAG results, reducing LLM API calls and improving response times. Cache keys use MD5 hashing with configurable TTLs.

TABLE II: Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 16, TailwindCSS, shadcn/ui	SSR React with modern UI
Backend	FastAPI, Python 3.11+, SQLAlchemy 2.0	Async API with 15 routers
LLM	Google Gemini 2.0 Flash (multimodal)	Reasoning, assessment, image analysis
Orchestration	LangGraph	Stateful multi-agent routing
Vector DB	Qdrant	Semantic search for RAG
Database	PostgreSQL 16	14 tables, 3 migrations
Cache	Redis 7	Query caching, analytics
Real-time	WebSocket + SSE	Study rooms + chat streaming
PDF	ReportLab	Progress report generation
Embeddings	all-MiniLM-L6-v2 (local)	384-dim sentence embeddings

IV. Agent Design

A. LangGraph Orchestrator: The orchestrator implements a StateGraph with nodes for state loading, intent classification, conditional routing to six agent nodes plus a fallback, state commitment, and response formatting.

Intent classification uses Gemini to map user messages to one of seven categories: conceptual_doubt, exam_prep, weak_points, teach_me, viva, critique, or general.

B. Inter-Agent Collaboration: Before routing to the selected agent, the streaming service enriches the agent state with cross-agent intelligence. Concept QA receives prerequisite gap alerts from the Weak-Point Tracker's data. Exam Predictor receives a list of mastered concepts to skip. Protégé and Viva agents receive weak area probing hints. This inter-agent context is injected into the LLM prompt as a "Cross-Agent Intelligence" section, enabling contextually aware responses without explicit agent-to-agent API calls.

C. Concept QA Agent: Retrieves relevant course material via RAG, identifies the student's confidence on the queried concept, checks for prerequisite gaps, and adapts explanation depth accordingly. Uses LLM-based assessment (not length heuristics) to evaluate student comprehension, comparing answers against RAG ground truth.

D. Exam Predictor Agent: Analyses syllabus coverage and student mastery to predict high-probability exam topics. Generates practice questions and, when students answer them, evaluates responses using the LLM against course material, closing the practice-feedback loop.

E. Weak-Point Tracker Agent: Monitors SM-2 confidence scores, identifies concepts below threshold, detects overdue reviews, and generates prioritised study recommendations. Integrates with the nudge engine to proactively surface review reminders.

F. Protégé Agent (Feynman Technique): The student teaches the AI, which plays a naive learner asking probing questions. After multiple exchanges, the agent provides structured evaluation with rubric scoring (clarity, accuracy, depth, examples) and updates concept confidence based on teaching quality.

G. Oral Viva Agent: Simulates viva-voce examination with adaptive difficulty (foundational → intermediate → advanced). Each question receives structured evaluation. Supports voice mode via Web Speech API for authentic oral exam experience.

H. Essay Critique Agent: Provides structured academic writing feedback with section-level rubric scoring (thesis, evidence, analysis, organisation, language), fact-checking against RAG context, and citation gap identification.

TABLE I: EduAgent Specialised Agents

Agent	Function	Key Technique
Concept QA	Answer conceptual doubts with RAG context	LLM-based assessment vs ground truth
Exam Predictor	Generate practice questions, evaluate answers	Mastery-aware question generation
Weak-Point Tracker	Surface weak/overdue concepts	SM-2 scheduling + confidence decay
Protégé	Student teaches AI (Feynman technique)	Multi-turn probing + rubric evaluation
Oral Viva	Simulate oral examination with voice	Difficulty ladder + Web Speech API
Essay Critique	Structured writing feedback	Section-level rubric + citation checking

V. Adaptive Learning Engine

A. SM-2 Spaced Repetition: EduAgent implements the SM-2 algorithm for scheduling concept reviews. Each StudentConcept record maintains easiness_factor, interval_days, and repetition count. Quality ratings (1-5) from agent assessments drive the algorithm, with confidence scores mapped as quality/5.

B. Auto-Generated Flashcards: After Concept QA conversations, the system uses Gemini to extract 3-6 key fact pairs (question/answer) and stores them as Flashcard records with independent SM-2 scheduling. Students can also create manual flashcards. The review interface presents cards with spaced repetition buttons (Forgot/Hard/Good/Easy) that drive the SM-2 update cycle.

C. Confidence Decay: Concepts not reviewed by their scheduled date undergo automatic confidence decay proportional to the overdue period. This ensures the system reflects realistic knowledge retention curves.

D. Smart Study Planner: Given an exam date and daily study hours, the planner queries the student's mastery summary, weak concepts, overdue reviews, and priority queue. It generates a day-by-day schedule using Gemini, allocating time blocks for review, practice, and teach-back activities. Plans are persisted for tracking.

E. Concept Dependency Auto-Detection: When course PDFs are uploaded, the system can automatically infer prerequisite relationships between existing concepts by analysing the content with Gemini. This builds the concept graph without manual edge creation by professors.

VI. Collaboration and Real-Time Features

A. WebSocket Study Rooms: Students can create or join study rooms for collaborative learning. The WebSocket endpoint authenticates via JWT tokens and maintains per-room connection pools. Messages are broadcast to all room members. Students can invoke the AI assistant directly within the room by sending an ai_question message type, with the response shared with all participants.

B. Voice Mode: The Oral Viva agent supports voice interaction through the Web Speech API. Speech-to-text captures student responses (SpeechRecognition), and text-to-speech (SpeechSynthesis) reads the AI's questions aloud. This creates an authentic oral examination experience without any external API costs.

C. Multimodal Image Understanding: Students can upload diagrams, flowcharts, or handwritten notes. The image is sent to Gemini's multimodal endpoint alongside the student's question. The system analyses visual content, explains components, identifies errors in diagrams, and transcribes handwritten text.

TABLE III: Key Features Summary

Feature	Description
Inter-Agent Collaboration	Agents consult weak points, mastery, overdue data before responding
Auto-Generated Flashcards	LLM extracts Q/A pairs from conversations, SM-2 review scheduling
Smart Study Planner	AI-generated day-by-day schedules based on mastery + exam dates
Voice Mode	Web Speech API for speech-to-text input and text-to-speech output
Image Understanding	Gemini multimodal analysis of diagrams and handwritten notes
Study Groups	WebSocket rooms with shared AI assistant
Custom Agent Builder	Admin-defined agents with custom prompts and rubrics
Class Analytics	Aggregated mastery, weak concepts, agent performance dashboards
Progress PDF Export	Downloadable mastery reports with tables and charts
Learning Path Timeline	Chronological visualisation of learning events
Concept Auto-Detection	LLM infers prerequisite edges from PDF content
Redis Caching	Reduced API calls and faster analytics responses

VII. Analytics and Observability

A. Agent Performance Logging: Every agent invocation is logged with agent type, intent classification, response time (milliseconds), RAG chunks used, and success status. The AgentLog table enables analysis of agent usage patterns, response time distributions, and error rates.

B. Class-Level Analytics: Administrators access aggregated dashboards showing class-wide mastery averages per subject, top weak concepts with student struggle counts, enrolled student counts, and total interaction volumes.

This enables data-driven curriculum adjustment.

C. Learning Path Timeline: Each significant learning event (concept learned, reviewed, mastered, weak point identified) is persisted as a LearningEvent record. The frontend renders these chronologically as an interactive timeline, providing visual storytelling of the student's learning journey.

D. Progress Report PDF Export: Students can download a professional PDF report generated by ReportLab containing their mastery summary, chapter breakdown with tables, weak points with priority tags, study priority queue, and overdue reviews.

E. Redis Caching: Analytics endpoints cache results for 5 minutes to avoid expensive database aggregations on every request. Cache keys are MD5-hashed and automatically invalidated on data changes.

VIII. Custom Agent Builder

Professors can create custom assessment agents through an admin interface. Each custom agent has a name, description, system prompt, and optional rubric (stored as JSONB). Examples include a 'Lab Report Reviewer' that checks for methodology and results analysis, or a 'Case Study Analyser' that evaluates problem-solving approach. Custom agents use the same base infrastructure (RAG, streaming, assessment) as built-in agents, providing extensibility without code changes.

IX. Implementation

The backend is implemented in Python 3.11+ with FastAPI, comprising 15 API routers, 9 service modules, 6 agent classes, and 14 database models. The frontend uses Next.js 16 with TypeScript, featuring 19+ pages and 15+ components.

Authentication uses JWT tokens with bcrypt password hashing, NextAuth.js on the frontend, and role-based access control (student/admin). The API provides auto-generated OpenAPI documentation at /docs.

The system is containerised with Docker Compose, running five services: PostgreSQL 16, Qdrant (vector DB), Redis 7 (cache), FastAPI backend, and Next.js frontend. Production and development compose files are provided, with multi-stage Dockerfiles for optimised images.

Testing includes 60+ automated tests: unit tests for SM-2 algorithm and password hashing, integration tests for intent classification, student state management, nudge engine, dashboard endpoints, orchestrator graph structure, and API endpoint validation. Frontend tests use Vitest with React Testing Library.

X. Evaluation

A structured user study protocol has been designed with 5 guided tasks (Concept Q&A;, Practice Questions, Weak Point Analysis, Protégé Mode, Dashboard Review) and a 21-item Likert-scale questionnaire covering usability, educational effectiveness, visualisation quality, agent quality, and comparison with alternatives. The System Usability Scale (SUS) is included for standardised benchmarking.

Quantitative metrics tracked include: agent response times (via AgentLog), intent classification accuracy, concept mastery improvement over time, flashcard retention rates, and study plan adherence. These metrics are accessible through the admin analytics dashboard.

[User study results to be inserted after conducting the study with 5-10 participants. Report mean Likert scores with standard deviations per section, SUS score vs. benchmark (68), and selected qualitative quotes from open-ended responses.]

XI. Discussion

EduAgent demonstrates that a multi-agent architecture with inter-agent collaboration provides a richer educational experience than single-agent chatbots. The key insight is that different learning activities (explanation, practice,

self-assessment, review) benefit from specialised agent behaviours and prompting strategies.

The inter-agent collaboration mechanism — where Concept QA considers weak points, Exam Predictor skips mastered topics, and Viva probes known weaknesses — represents a lightweight but effective form of agent communication. Rather than direct agent-to-agent API calls, we inject cross-agent context into prompts, allowing the LLM to naturally incorporate this awareness.

Limitations include: (1) dependency on a single LLM provider (Gemini), creating a single point of failure; (2) the SM-2 algorithm's simplicity compared to more sophisticated memory models; (3) lack of real-time collaboration features beyond text (e.g., shared whiteboards); and (4) the need for larger-scale user studies to validate learning outcome improvements.

Future work includes: fine-tuning smaller models for intent classification to reduce latency, implementing learning outcome prediction models, adding collaborative whiteboarding, supporting multiple LLM backends, and conducting longitudinal studies measuring actual grade improvements.

XII. Conclusion

This paper presented EduAgent, a multi-agent AI educational platform that goes beyond single-chatbot solutions by providing six specialised agents with inter-agent collaboration, sharing a unified knowledge layer. The system integrates SM-2 spaced repetition, auto-generated flashcards, AI-powered study planning, voice-enabled oral examination, multimodal image understanding, collaborative study rooms, custom agent building, and comprehensive analytics.

The platform is implemented as a production-ready full-stack application with 60+ automated tests, five-service Docker deployment, and extensive documentation. The combination of adaptive learning algorithms, multi-agent AI, and real-time collaboration features demonstrates the potential for AI-driven educational tools that genuinely adapt to individual student needs.

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